Universität Konstanz

Introduction to Computing for the Social Sciences

Session 13 – Social Science Applications

Question & answer examples

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Three questions for this session

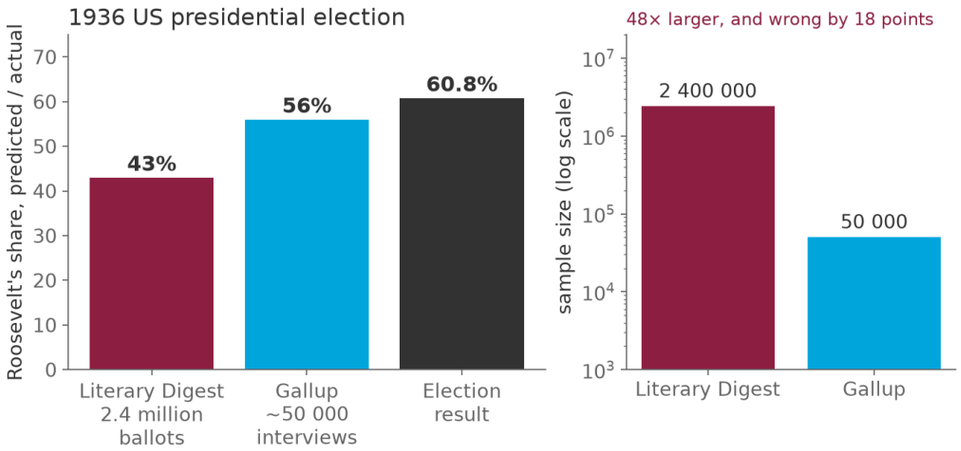
Each example is an **opening** question for the start of the lecture and an **answer** for the end, followed by a notes slide. Copy the slides you want into the main deck.

	Question	Hook	id
A	Two point four million ballots, and the wrong president	The Literary Digest, 1936: the biggest poll in history, and why size did not save it.	`X13-A`
B	Is every segregated city full of racists?	A grid, a rule, and a result that separates outcomes from motives.	`X13-B`
C	How many Likes does it take to know you better than your spouse?	A number, a method, and a privacy problem that consent cannot fix.	`X13-C`

Main deck `Slides/13-SocialScienceApplications/` · transcript `Transcripts/13-…md` (the § in the answers)

A · Two point four million ballots, and the wrong president

Opening question · place after the Overview slide



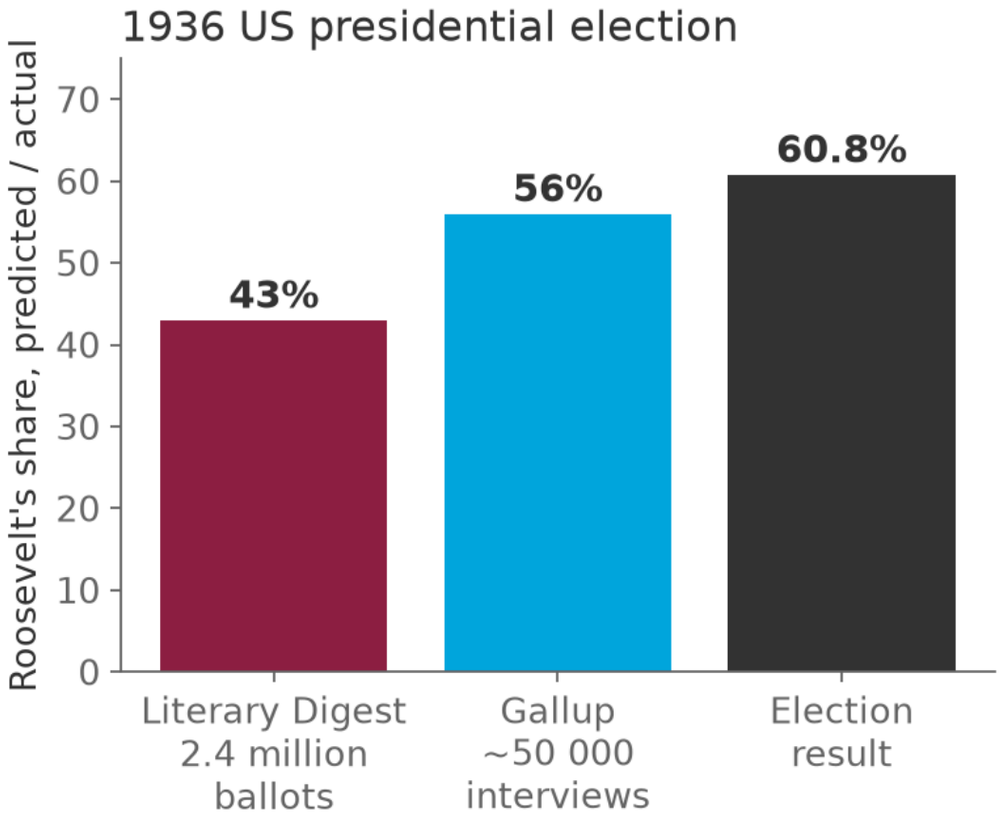
In 1936 the *Literary Digest* mailed 10 million ballots and received **2.4 million** back, the largest poll ever taken. It predicted **Landon 57%, Roosevelt 43%**. A young pollster named George Gallup, with about **50 000** interviews, predicted Roosevelt.

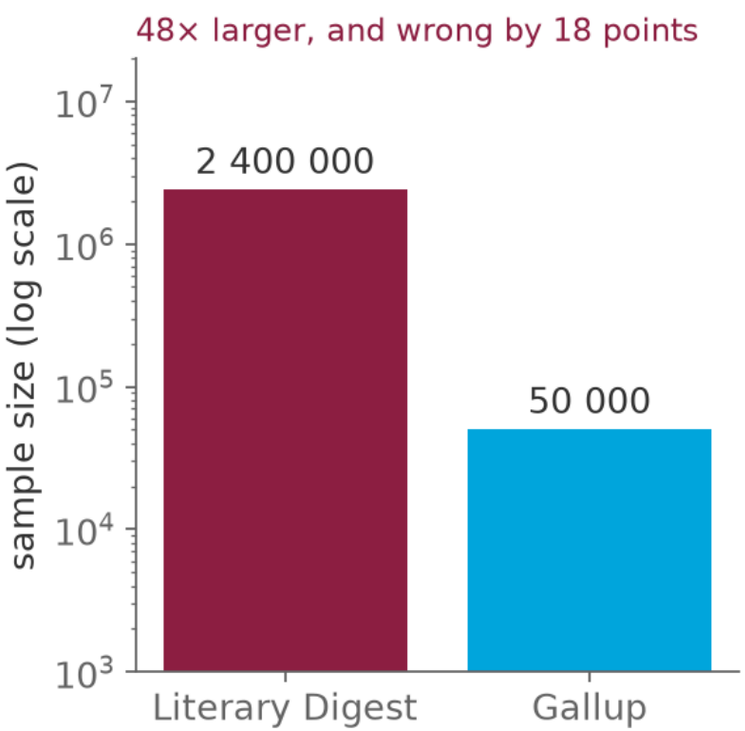
Roosevelt won with **61%**. The *Digest* folded within two years.

How can 2.4 million responses be wrong by 18 points while 50 000 are right? And what does that say about a study of 500 million tweets?

A · Two point four million ballots, and the wrong president · answer

The answer · place before the Summary slide





A · Two point four million ballots, and the wrong president · why

The answer · place before the Summary slide

- The *Digest* mailed to **its subscribers, telephone owners and car owners**, in 1936 the affluent. **Coverage error**: the sampling frame was not the electorate. The 24% who answered were also the most motivated: **non-response bias**.
- Gallup's sample was small but **designed to represent** the population.
- Size reduces **variance**, never **bias**. Digital trace data is the *Digest* again: a huge, self-selected, platform-shaped sample. The question is always *who is in the frame*, not *how many*.

What we used: §6 digital behavioral data · §8 issues with representation: coverage, behavioral distortion, algorithmic confounding · §12 why data does not replace theory

A · Two point four million ballots, and the wrong president · notes

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X13-A

Source. Squire, P. (1988), “Why the 1936 Literary Digest Poll Failed”, *Public Opinion Quarterly* 52(1), which shows that non-response mattered at least as much as the frame. Actual result: Roosevelt 60.8%, Landon 36.5%. Gallup’s sample size varies by account (50 000 is the commonly cited figure).

Facts. 10 million ballots mailed, about 2.4 million returned (24%). The *Digest* had correctly predicted every election since 1916. Squire’s re-analysis: even among the mailed frame, respondents leaned Landon more than non-respondents.

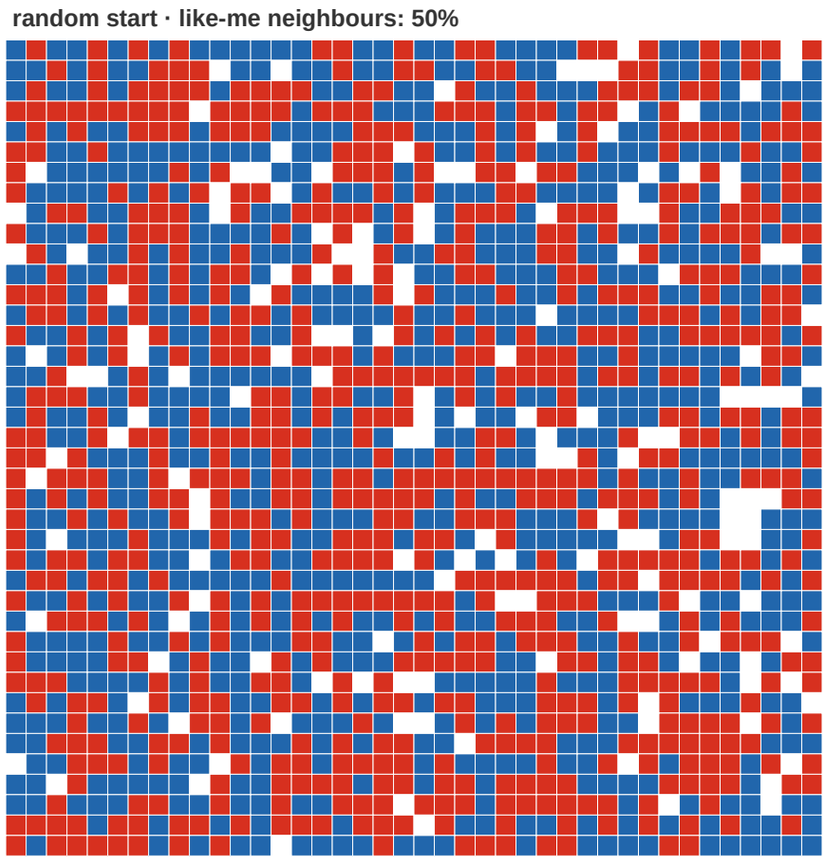
How to present. Open with the three bars and “which do you believe?”; the room will pick the big one. Close with the two mechanisms named and separated (frame against response), then the transfer to platforms: the Pew age-gap table from the main deck is the modern coverage error. The line to land: **a biased sample of a billion is a biased sample**.

Where it plugs in. Before the “Issues with digital behavioral data: representation” slide of the main deck.

Self-study: post `Examples/posts/X13-A-….md` · notebook `Examples/notebooks/X13-A.ipynb`, the problem and the solution in a few cells.

B · Is every segregated city full of racists?

Opening question · place after the Overview slide



1 440 agents of two kinds, placed at random on a 40×40 grid with 10% of cells empty. Each agent wants **at least 30% of its neighbors** to be like itself; it is content to be in a **70% minority**.

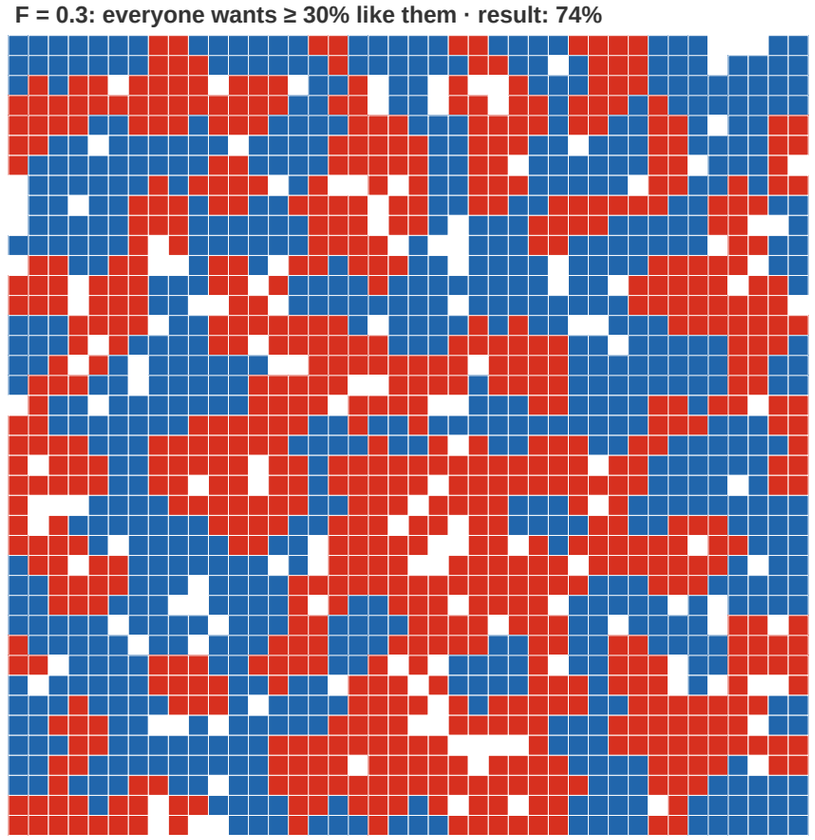
Unsatisfied agents move to a free cell where they would be satisfied.

What does the grid look like when everyone is satisfied? And if you saw only the end state, what would you conclude about the agents' preferences?

simulated with `tools/genexamples_figs.py`

B · Is every segregated city full of racists? · answer

The answer · place before the Summary slide



The question was: What does the grid look like when everyone is satisfied? And if you saw only the end state, what would you conclude about the agents' preferences?

simulated with `tools/genexamples_figs.py`

B · Is every segregated city full of racists? · why

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The answer · place before the Summary slide

- After **8 rounds** and 384 moves everyone is satisfied, and the grid is strongly segregated: the neighborhood of the average agent went from **50%** to **74%** like itself, far above the 30% anyone asked for.
- Each move changes the neighborhoods of the cells left and joined, tipping others below threshold. Mild local preferences **compound** into global sorting.
- **Generative sufficiency**: a segregated outcome does not imply intolerant preferences. You cannot read motives off aggregates. This is the model's point, and the reason agent-based modeling belongs in the toolkit.

What we used: §13 agent-based modeling and generative sufficiency · §14 Schelling's segregation model · §3 the 'generative' meaning of computational

B · Is every segregated city full of racists? · notes

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X13-B

Source. Schelling, T. C. (1971), “Dynamic Models of Segregation”, *J. Math. Sociology* 1(2). The two grids are produced by a real run of the model in the figure script (seed 11, $F = 0.3$, 10% empty, 40×40); the 50% to 74% and “8 rounds” figures are read from that run and change together if regenerated.

Facts. Schelling’s original threshold examples are 1/3 and 1/2; the CMSS course slides run the full sweep. At $F = 0.3$ the result is already clear; at $F = 0.5$ it is stark. The measure shown is the mean share of same-kind neighbors among occupied neighboring cells.

How to present. Open with the random grid and the rule; ask for a sketch of the end state (most draw it *less* clustered than it turns out). Close with the after-grid and the two numbers; then the inference question. The exam item `DB-14` asks about the same model, with $F = 0.5$ in its part (e). Add: this is also the answer to *“why simulate?”*; the result is not derivable from the rule by inspection.

Where it plugs in. With the Schelling slides of the main deck (properties, F = 0.2 simulation, tolerance results).

Self-study: post `Examples/posts/X13-B-….md` · notebook `Examples/notebooks/X13-B.ipynb`, the problem and the solution in a few cells.

C · How many Likes does it take to know you better than your spouse?

Opening question · place after the Overview slide

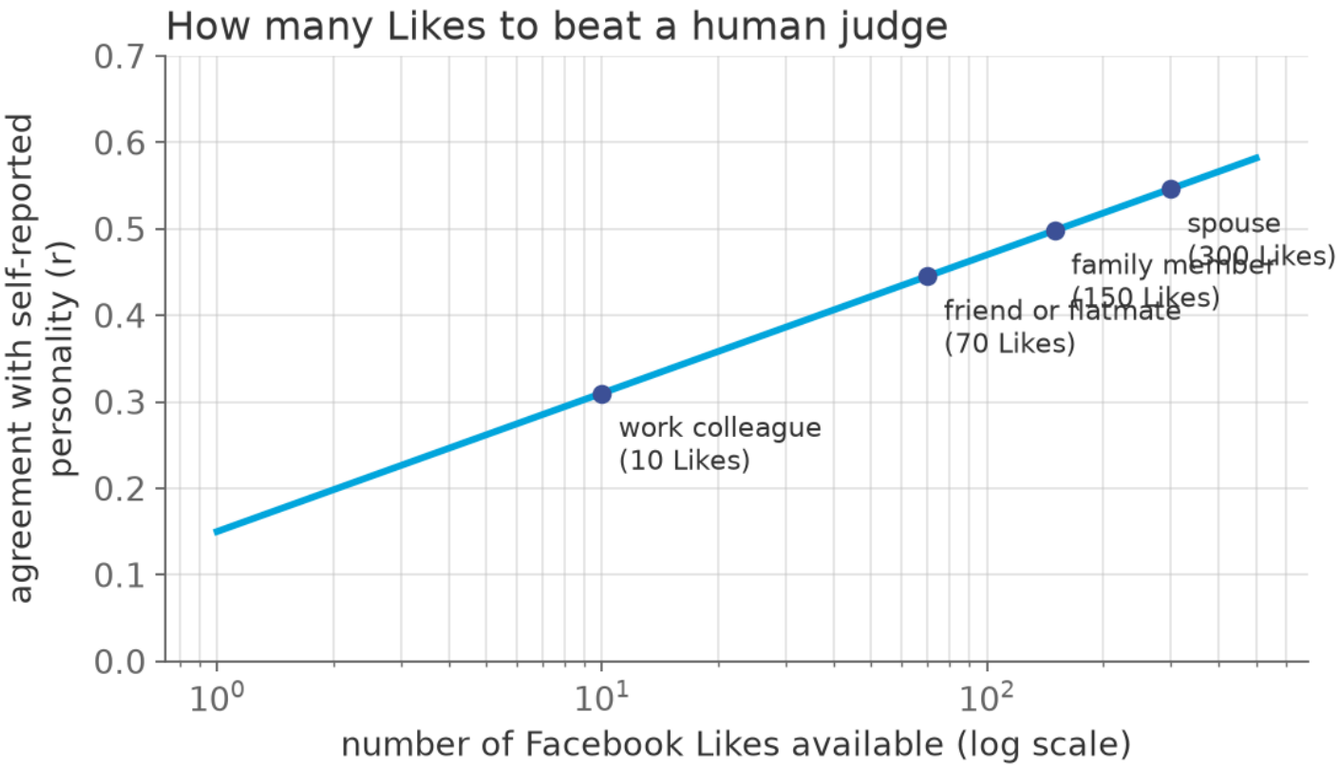
Personality, the Big Five, is normally measured with a 100-item questionnaire. In 2015 researchers built a model that predicts it from a person's **Facebook Likes** alone, and compared its accuracy to that of the person's friends, family and spouse, who each rated the same person.

The judges knew the person for years. The model had a list of Likes.

How many Likes did the model need to be as accurate as a work colleague? A friend? A spouse? And what follows for privacy, even for people who post nothing?

C · How many Likes does it take to know you better than your spouse? · answer

The answer · place before the Summary slide



schematic after Youyou, Kosinski & Stillwell (2015)

C · How many Likes does it take to know you better than your spouse? · why

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The answer · place before the Summary slide

- **10 Likes** to match a work colleague, **70** a friend, **150** a family member, **300** to match the spouse. The average user then had about 227.
- Each Like is innocuous; the **pattern** is not. Kosinski et al. (2013) predicted sexual orientation, religion, politics and substance use from Likes at high accuracy.
- **Shadow profiles**: your friends' Likes predict *you*. Privacy is **collective**; no individual opt-out removes the information others' data carries about you.

What we used: §10 method examples: personality from Likes · §11 issues with methods: privacy and shadow profiles · §7 digital traces as measurement · session 10 §3 supervised learning

C · How many Likes does it take to know you better than your spouse? · notes

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X13-C

Source. Youyou, W., Kosinski, M. & Stillwell, D. (2015), “Computer-based personality judgments are more accurate than those made by humans”, *PNAS* 112(4): 86 220 participants; thresholds 10/70/150/300 from the paper. Kosinski, Stillwell & Graepel (2013), *PNAS* 110(15). Garcia et al. (2018), *EPJ Data Science* 7:3, for shadow profiles. The curve in the figure is **schematic**: the anchors are from the paper, the line between them is drawn, and the slide says so.

Facts. Human judges: self-other agreement $r \approx 0.49$; computer with the average number of Likes $r \approx 0.56$. The model was a linear regression on the Like matrix, nothing exotic.

How to present. Open with the setup and ask for the four numbers; guesses are usually far too high for the colleague and too low for the spouse. Close with the anchored curve, then the two implications in the last two bullets; the second is the one students have not thought about. Connect to session 10 (this is plain supervised learning) and to the ethics list of session 10.

Where it plugs in. With the “Personality and Facebook likes” and “New privacy risks” slides of the main deck.

Self-study: post `Examples/posts/X13-C-….md` · notebook `Examples/notebooks/X13-C.ipynb`, the problem and the solution in a few cells.